

Ex:No: 8

Date:

Iris Flower Clustering Using K-Means Algorithm

Problem Description:

The Iris dataset contains 150 samples of three iris species (**Setosa, Versicolor, Virginica**) with four numerical features: **Sepal Length, Sepal Width, Petal Length, Petal Width**. The goal is to **group the flowers into clusters** based on these features using the **K-Means clustering algorithm**, without using the species labels.

Objective:

The objective of this program is to implement the K-Means clustering algorithm in R to group iris flowers based on their numerical attributes, visualize the clusters, and evaluate clustering performance against the actual species.

```
# Step 1: Install and Load Required Libraries
```

```
install.packages("factoextra")
```

```
install.packages("cluster")
```

```
install.packages("ggplot2")
```

```
library(factoextra)
```

```
library(cluster)
```

```
library(ggplot2)
```

```
# Step 2: Load and Prepare Dataset
```

```
data(iris)
```

```
# Remove Species column
```

```
iris_data <- iris[, -5]
```

```
# Display first few rows
```

```
head(iris_data)
```

```
# Step 3: Elbow Method to Find Optimal k
```

```
fviz_nbclust(iris_data, kmeans, method = "wss") +
```

```
  ggtitle("Elbow Method for Optimal k")
```

```
# Step 4: Apply K-Means Clustering (k = 3)
```

```
set.seed(123)
```

